What we have built so far, in plain words

UAV-NOMA Disaster Network Planning Simulator. Status on 22 October 2026. Full details: the project book (docs/book).

The idea in one paragraph

After a disaster, phone towers are often down. A drone can carry a small "flying phone tower" and give survivors a signal. Our project is a **computer simulation** of that situation. It answers: *where should the drone fly, at what height, and how should it share its signal, so that as many survivors as possible stay connected before its battery runs out?* The drone shares its signal using **NOMA**, a newer technique that lets two people use the same slice of radio at the same time. We compare simple, hand-written ways of flying the drone with an **AI that learns to fly by itself**.

What is built

Part	What it is
Disaster simulator	A virtual disaster area: up to 1,500 survivors in groups, collapsed buildings that block signals, people who walk or suddenly move, a drone with a realistic battery
Radio model	How strong the signal is at each person (from a published model), and how many people get a good enough connection with NOMA or with the older technique (OMA)
Dataset	Over 800,000 rows of results, also as an Excel file
8 programmed pilots (B0-B7)	Simple rules for flying the drone, from "stay parked in the middle" to "search every spot" and "circle to save battery"
The AI pilot	A program that learns to fly by trial and error, seeing only what a real drone could measure
Dashboard	A web page to set up a disaster, fly any pilot, compare them and get a recommended plan
Project book	14 chapters, a PDF, charts and animations; every decision recorded
Quality checks	315 automatic tests, run on every change on GitHub

What we have proven

1. A drone that moves beats a parked drone (hypothesis H1). When crowds move, a drone that follows them serves **74 of every 100 people**; a drone parked in the middle serves **43**.

2. NOMA beats the older technique (hypothesis H3). We gave both techniques the same smart tuning, to be fair. NOMA still serves **3 to 5 more people out of every 100**, in every situation we tried, and gives more total speed. In the easier scenarios it serves **98 of every 100 people**, and almost no rescue workers are left without signal (0.4%). It still wins when the phone cannot remove the other person's signal perfectly.

Chart: NOMA vs OMA, same tuning for both

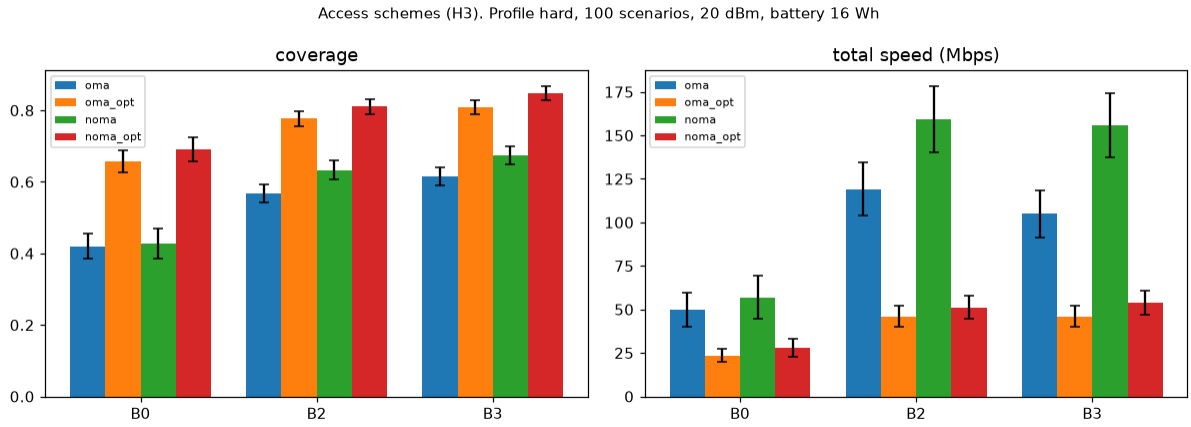


Figure: 8_access_schemes_hard.png

3. Circling saves battery. In our energy model a drone flying slow circles uses less power than one hovering in place. Circling serves **26% more people per battery charge**, and with a small battery it finishes the mission while a hovering drone runs out. This holds even when we change the uncertain energy numbers.

4. The AI learns by itself. Seeing only rough crowd maps (no exact positions), it went from worse than random to **0.576**, close to a hand-written pilot that knows more (0.610). It finds the best spot within 30 seconds.

What did not work, and what we learned

- **The AI does not stay at the best spot.** It finds the best spot and then flies past it. On standing crowds it ended about 0.03 below the hand-written pilot, so the first AI checkpoint (gate G4) was not passed. The fix ("move by" steering and longer training) is planned for the next stage.

Picture: the AI (green line) flies past the best spot; the programmed pilots stay

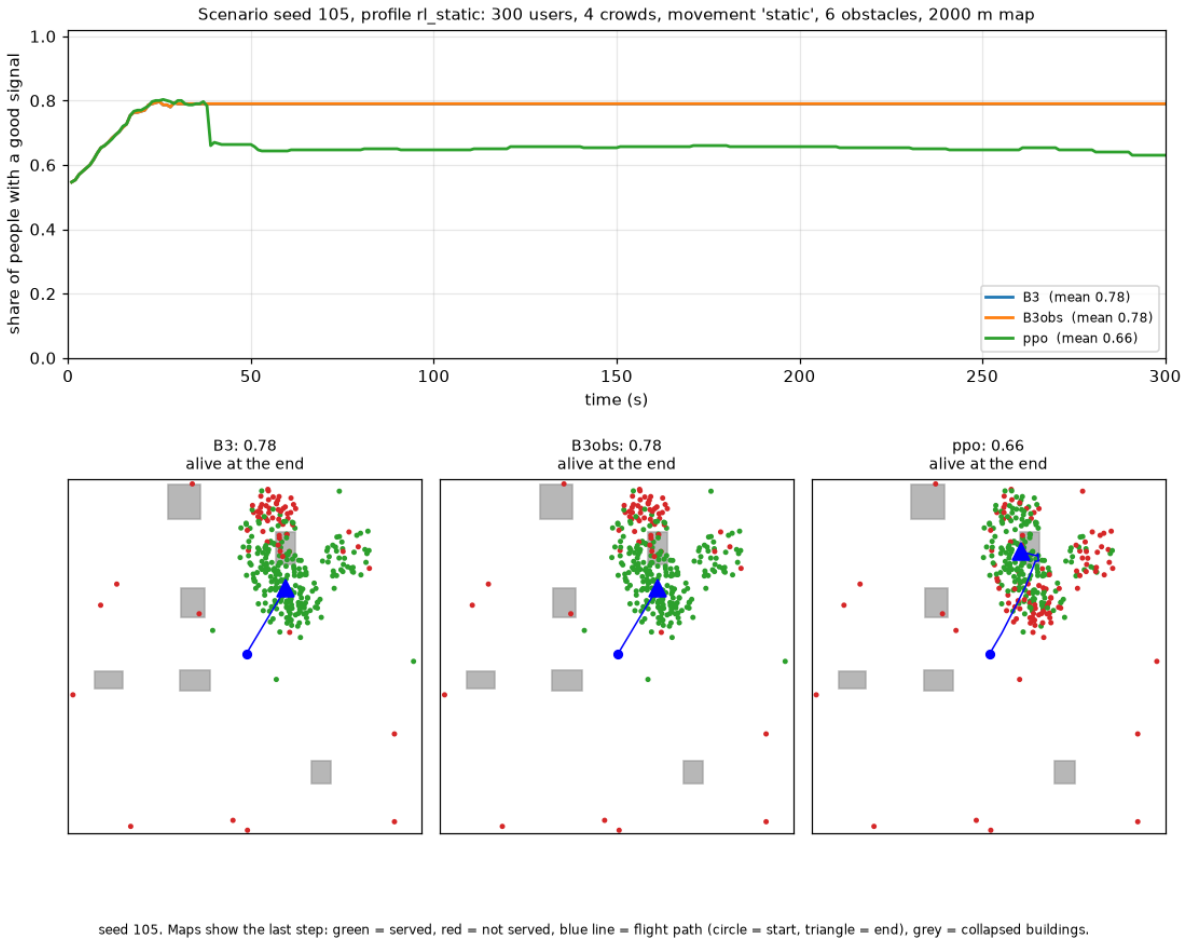


Figure: 10_e0_vs_b3_seed105.png

- **On walking crowds no AI can win.** We built a "fortune-teller" drone that knows where everyone will walk. Even it gains only 0.2% over a simple pilot, because the drone (15 m/s) is much faster than people walking (1-3 m/s). This saved us two weeks of useless training, and it is a finding in itself: for slow situations, a simple program is enough.
- **We found and closed a hidden flaw** in the simulator: the drone could "shake" on the spot to fake a battery saving. It is now fixed, and no earlier result had used it.

Where we are now: finding where an AI can win

We redesigned the scenarios so that **the need for help suddenly jumps between places far apart**: for example a phone tower is repaired at one site and a shelter opens at another, 1-2 km away. Results (12 scenarios per design, still to be confirmed on fresh scenarios): a drone that thinks ahead could serve **up to 11 more people out of every 100**, compared with 0.2 for walking crowds. The rule we wrote down beforehand (at least 2 more) is met in 7 of the 9 designs, most clearly when the sites are 2 km apart.

Distance between sites	Sudden switch	Switch over 1 minute	Switch over 3 minutes
1.0 km	**+2.5**	+0.9	+0.5
1.5 km	**+7.2**	**+3.9**	+2.0 (just)
2.0 km	**+10.8**	**+9.8**	**+5.6**

(More people served out of every 100 by a drone that knows what is coming, compared with the best simple pilot.)

What comes next

1. Finish this check, repeat it on fresh scenarios, and add the strongest simple pilot (one that also uses announced schedules).
2. **Train the AI on these new scenarios** (the main experiment, about two weeks).
3. Disaster events (rising flood, aftershock), then realism (wind, GPS errors), then the final evaluation and report.
4. Last: a realistic 3D simulated drone flying with our controller.

How we work

Every important plan is checked by an independent reviewer first. The rules for every experiment are written down **before** it runs, so the goalposts never move. Every result is reported, including the ones where the AI loses, and nothing is called "proven" unless the numbers clearly show it.